

7- Find the total order costs (price + shipping_charges) by city. List the city and the total

order cost. List only cities with order costs of 50 or less.

```
select sum(total_cost) as city_totoal_cost, customer_city
from (
select shipping_charges, price, (shipping_charges + price) as total_cost, customer_city
from orderitems join orders using (order_id) right join customers using (customer_id)
) as customer_cost
group by customer_city
having city_totoal_cost <= 50;
/* 7 rows returned */
```

	city_totoal_cost	customer_city
▶	7	ibatiba
	24.73	corguinho
	36.38	witmarsum
	44.019999999999996	rio claro
	38.989999999999995	brusque
	32.489999999999995	botucatu
	40.49	pindamonhangaba

8- Find product categories that have more than the average number of products across all categories. Hint: You need to use a subquery.

```
select count(product_id) as num_products, product_category_name from products
group by product_category_name
having num_products > (
select avg(num_products) from (
select count(product_id) as num_products, product_category_name from products
group by product_category_name
) as category_num);
/* 4 rows returned */
```

	num_products	product_category_name
▶	25	health_beauty
	726	toys
	26	bed_bath_table
	32	sports_leisure

9- List customer cities where the average price of orders from their citizens exceeds 5 times the average price of all orders. Hint: Only consider price.

```
select avg(price), customer_city
```

```

from orderitems join orders using(order_id) right join customers using(customer_id)
group by customer_city
having avg(price) > 5*(select avg(price) from orderitems);
/* 11 rows returned */

```

	avg(price)	customer_city
▶	1590	praia grande
	2799.65	mesquita
	1815.91	colombo
	1653.88	santarem
	1899	carmo do paranaiba
	2130	volta redonda
	1589	nudeo residencial pilar
	1849	piedade
	1850	sao miguel do araguaia
	2899	alagoa

10- List the top 5 customers who placed the most orders (in terms of total price) among all orders, calculate their total number of orders and total amount spent, and sort them in descending order by total amount they have spent. Only consider price as order cost.

```

select sum(price) as total_amount, count(order_id) as num_orders, customer_id
from orderitems join orders using(order_id) group by customer_id
order by total_amount desc
limit 5;
/* 5 rows returned */

```

	total_amount	num_orders	customer_id
▶	2899	1	04ToBIcjcnSO
	2899	1	00hixzlyOIta
	2899	1	0BWZV6c8XBVF
	2799.65	1	03eROJE4EsnZ
	2218	1	0aUDXPU65z9t

11- Find the active sellers who have at least sold 5 products.

```

select seller_id, count(product_id) as products_sold
from orderitems group by seller_id having products_sold >= 5;
/* 15 rows returned */

```

	seller_id	products_sold
▶	ruq0u2ZpAMDr	5
	SgeUigb5Vbiv	6
	yHsOFiqVrV1u	6
	K0qPVGdA91KO	10
	9ffFM6RBxCkT	5
	l1pYW6GBnPMr	5
	RKad98cTxhSb	12
	UOGIrJtSldvd	10
	coi9xkdo66JI	9
	7s071shkzUlk	5

12- Find the average shipping charge and number of orders per customer city. List only those who had more than 2 orders. Sort by average shipping charges in decreasing order.

```
select avg(shipping_charges), count(order_id) as num_orders, customer_city
from orderitems join orders using(order_id) join customers using (customer_id)
group by customer_city
having num_orders > 2
order by avg(shipping_charges) desc;
/* 30 rows returned */
```

	avg(shipping_charges)	num_orders	customer_city
▶	137.83249999999998	4	contagem
	109.98666666666668	3	macae
	100.27333333333333	3	jundiai
	73.0425	4	goiania
	72.922	5	recife
	60.383333333333326	3	vinhedo
	59.89	5	guarulhos
	52.47666666666667	3	sete lagoas
	51.32249999999999	4	campo grande

13- List the products that are heavier than the average weight of all products.

```
select * from products
where product_weight_g > (select avg(product_weight_g) from products);
/* 213 rows returned*/
```

	product_id	product_category_name	product_weight_g	product_length_cm	product_height_cm	product_width_cm
▶	00FVdl7l7Lz	toys	3300	31	32	31
	00hf4DIVWtOv	toys	2775	29	27	18
	00j9JICga8vf	toys	19783	49	54	52
	0133fSLGcAdY	toys	9250	56	11	32
	02CURpoL9IPa	toys	1900	33	27	26
	02HHGHzbG7T5	toys	3100	30	11	30
	02sfxo5wNODr	toys	5000	50	45	25
	03SwknUmJAW3	toys	11825	64	25	40
	04AxXojGGpj	toys	5000	42	16	37

14- How much time did it take for you to finish?

1 hour